

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

Data Preview

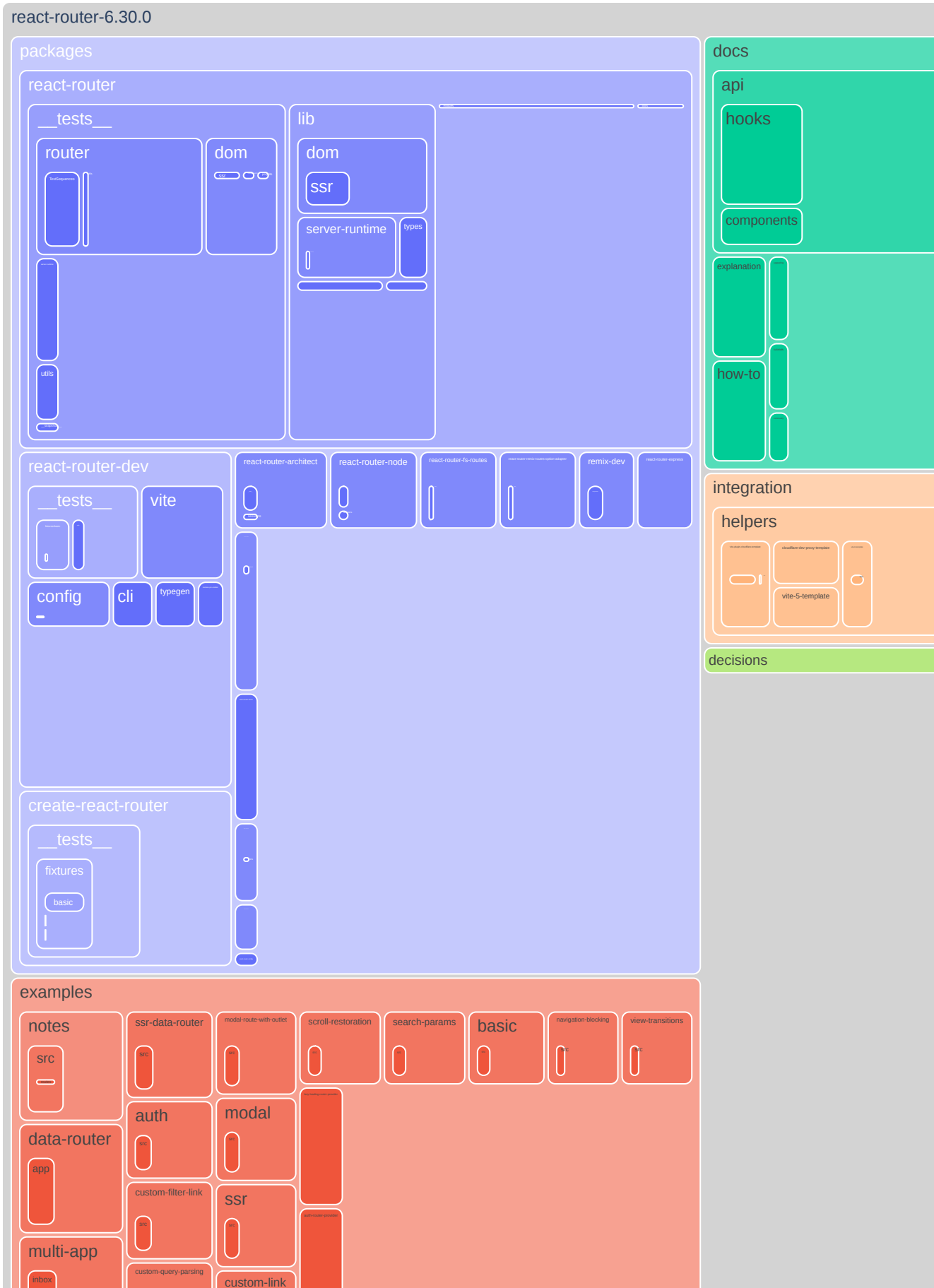
	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	203.000000	203.000000	203.000000	203.000000	203.000000	203.000000
mean	22.305419	13.300493	653.871921	117.236453	397.443350	180.620690
std	100.597499	38.987601	3261.376257	221.791342	469.421427	303.413532
min	1.000000	2.000000	2.000000	0.000000	0.000000	0.000000
25%	2.000000	3.000000	21.500000	10.000000	76.500000	9.000000
50%	5.000000	6.000000	63.000000	27.000000	158.000000	34.000000
75%	13.000000	10.000000	253.000000	136.000000	627.000000	158.000000
max	1296.000000	487.000000	38110.000000	1251.000000	1770.000000	1304.000000

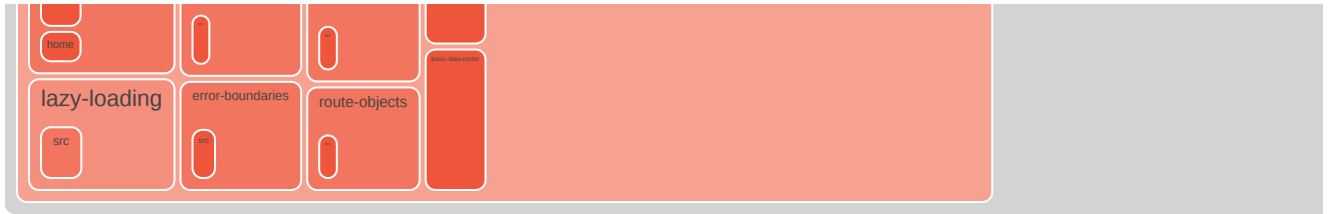
	directoryPath	directoryParentPath	directoryName	fileCount	firs
0	react-router-6.30.0/examples/custom-query-parsing/types	react-router-6.30.0/examples/custom-query-parsing	types	1	react-router-6.30.0/examples/custom-query-parsing/types/jsur
1	react-router-6.30.0/examples/data-router/src	react-router-6.30.0/examples/data-router	src	1	react-router-6.30.0/examples/data-router/src/la
2	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app	routes	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes/_inde
3	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template	public	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public/favico
4	react-router-6.30.0/integration/helpers/vite-5-template/app/routes	react-router-6.30.0/integration/helpers/vite-5-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-5-template/app/routes/_inde
5	react-router-6.30.0/integration/helpers/vite-5-template/public	react-router-6.30.0/integration/helpers/vite-5-template	public	1	react-router-6.30.0/integration/helpers/vite-5-template/public/favico
6	react-router-6.30.0/integration/helpers/vite-6-template/app/routes	react-router-6.30.0/integration/helpers/vite-6-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-6-template/app/routes/_inde
7	react-router-6.30.0/integration/helpers/vite-6-template/public	react-router-6.30.0/integration/helpers/vite-6-template	public	1	react-router-6.30.0/integration/helpers/vite-6-template/public/favico
8	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes/_inde
9	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	public	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public/favico
10	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	workers	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers/a
11	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes/hom
12	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public/favico
13	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	blank	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank/package
14	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	with-ignored-dir	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir/package
15	react-router-6.30.0/packages/react-router-architect/sessions	react-router-6.30.0/packages/react-router-architect	sessions	1	react-router-6.30.0/packages/react-router-architect/sessions/arcTableSessionStora
16	react-router-6.30.0/packages/react-router-cloudflare/sessions	react-router-6.30.0/packages/react-router-cloudflare	sessions	1	react-router-6.30.0/packages/react-router-cloudflare/sessions/workersKVStora
17	react-router-6.30.0/packages/react-router-config	react-router-6.30.0/packages	react-router-config	1	react-router-6.30.0/packages/react-router-config/LIC
18	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes/_inde
19	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public/favico
20	react-router-6.30.0/packages/react-router-dev/vite/static	react-router-6.30.0/packages/react-router-dev/vite	static	1	react-router-6.30.0/packages/react-router-dev/vite/static/refresh-uti
21	react-router-6.30.0/packages/react-router-dom/docs	react-router-6.30.0/packages/react-router-dom	docs	1	react-router-6.30.0/packages/react-router-dom/docs/README
22	react-router-6.30.0/packages/react-router-express/__tests__	react-router-6.30.0/packages/react-router-express	__tests__	1	react-router-6.30.0/packages/react-router-express/__tests__/server-ti
23	react-router-6.30.0/packages/react-router-native/docs	react-router-6.30.0/packages/react-router-native	docs	1	react-router-6.30.0/packages/react-router-native/docs/README
24	react-router-6.30.0/packages/react-router-node/sessions	react-router-6.30.0/packages/react-router-node	sessions	1	react-router-6.30.0/packages/react-router-node/sessions/fileStora
25	react-router-6.30.0/packages/react-router/__tests__/__snapshots__	react-router-6.30.0/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.0/packages/react-router/__tests__/__snapshots__/route-matc test.tsx
26	react-router-6.30.0/packages/react-router/__tests__/dom/components	react-router-6.30.0/packages/react-router/__tests__/dom	components	1	react-router-6.30.0/packages/react-router/__tests__/dom/components/LazyComponer
27	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills	react-router-6.30.0/packages/react-router/__tests__/dom	polyfills	1	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills/drop-Forml submit
28	react-router-6.30.0/packages/react-router/docs	react-router-6.30.0/packages/react-router	docs	1	react-router-6.30.0/packages/react-router/docs/README

	directoryPath	directoryParentPath	directoryName	fileCount	firs
29	react-router-6.30.0/playground/framework-express/app/routes	react-router-6.30.0/playground/framework-express/app	routes	1	react-router-6.30.0/playground/framework-express/app/routes/_inde

Number of files per directory

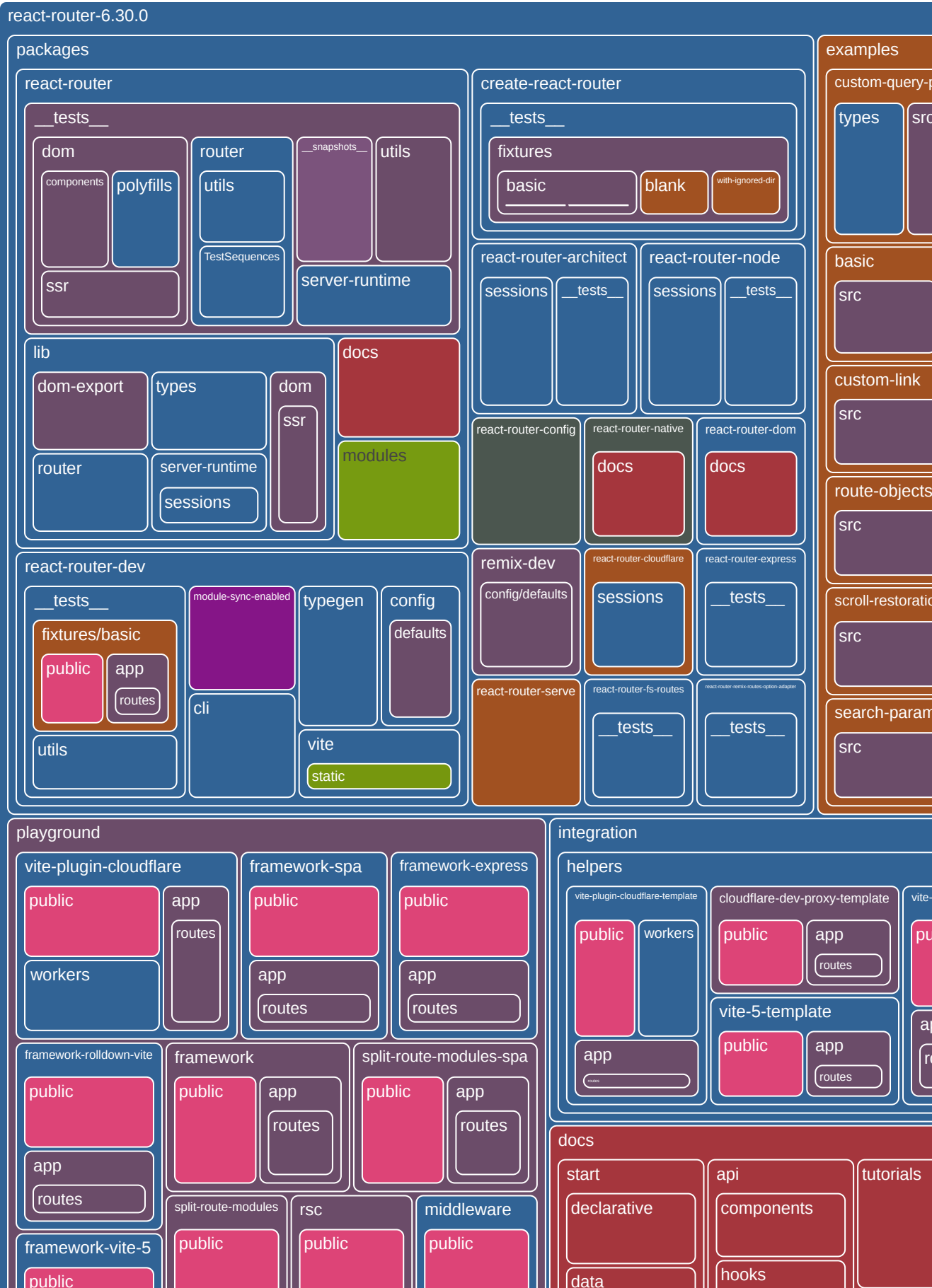
Directories and their file count

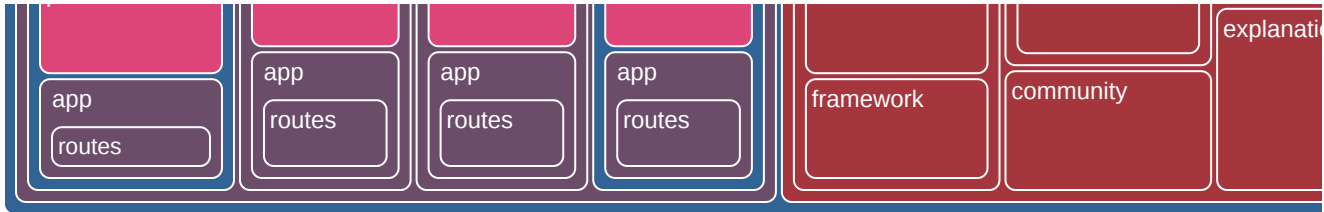




Most frequent file extension per directory

Most frequent file extension per directory

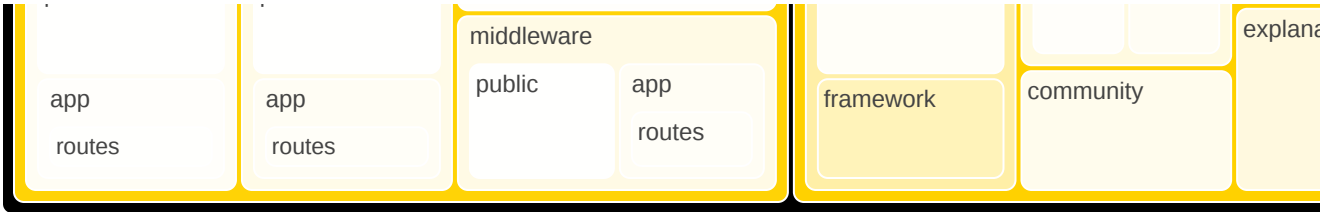




Number of commits per directory

Number of git commits

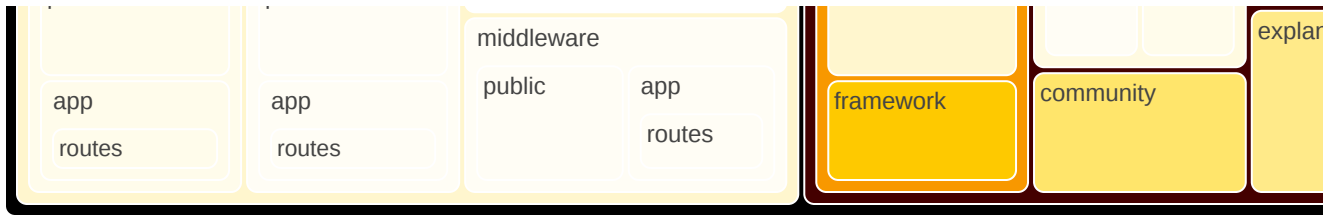




Number of distinct authors per directory

Number of distinct commit authors



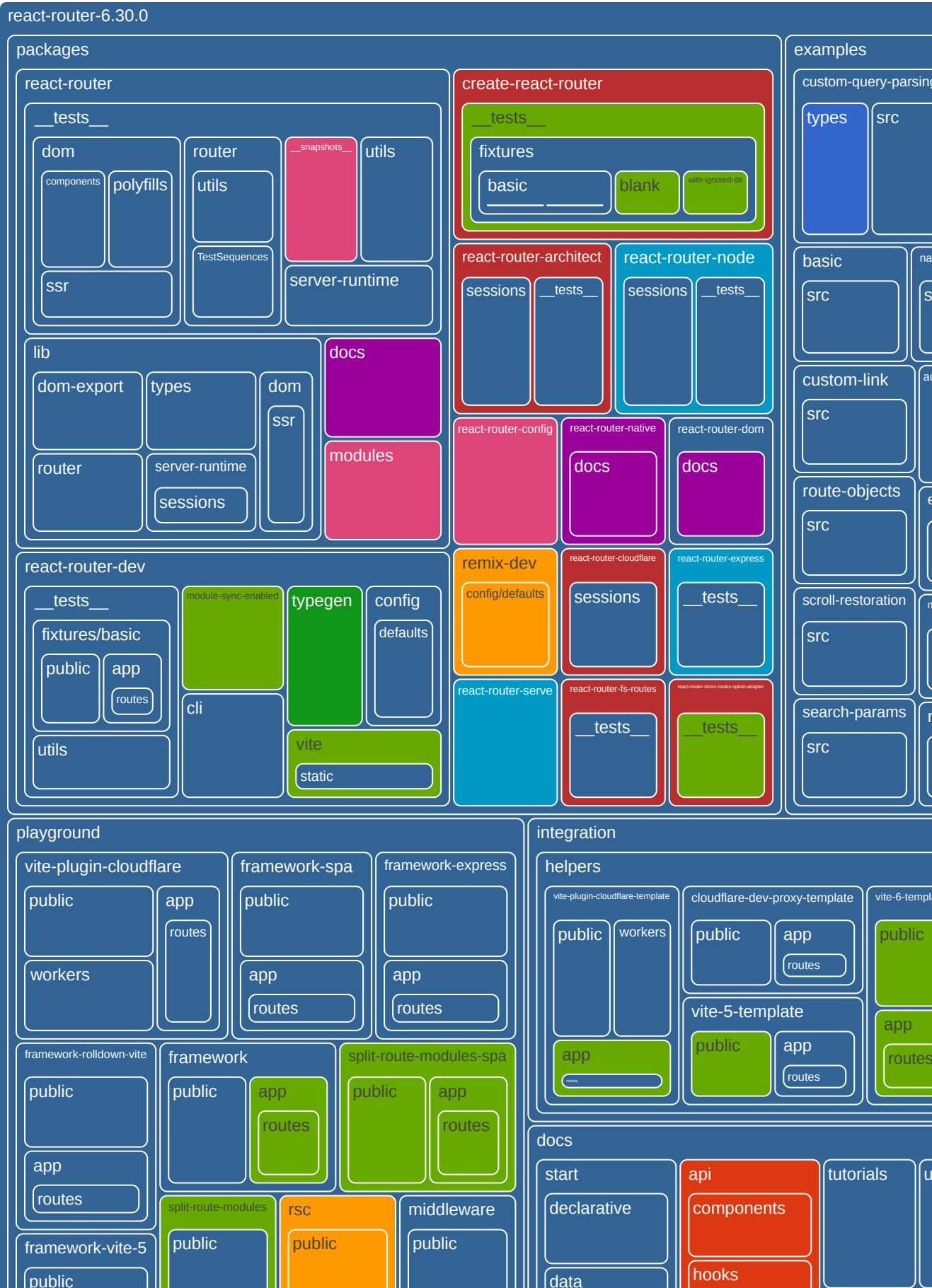


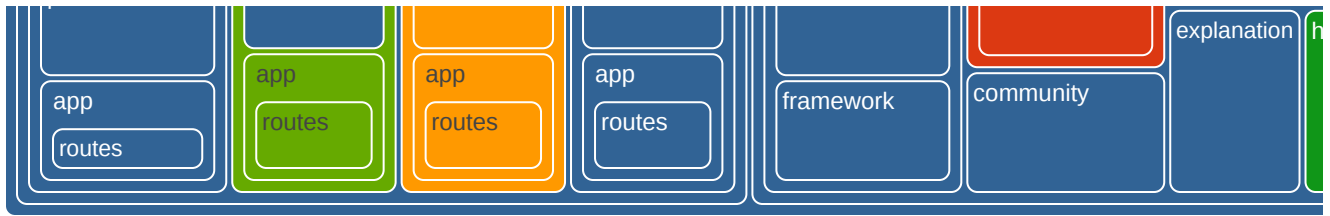
Directories with very few different authors



Main author per directory

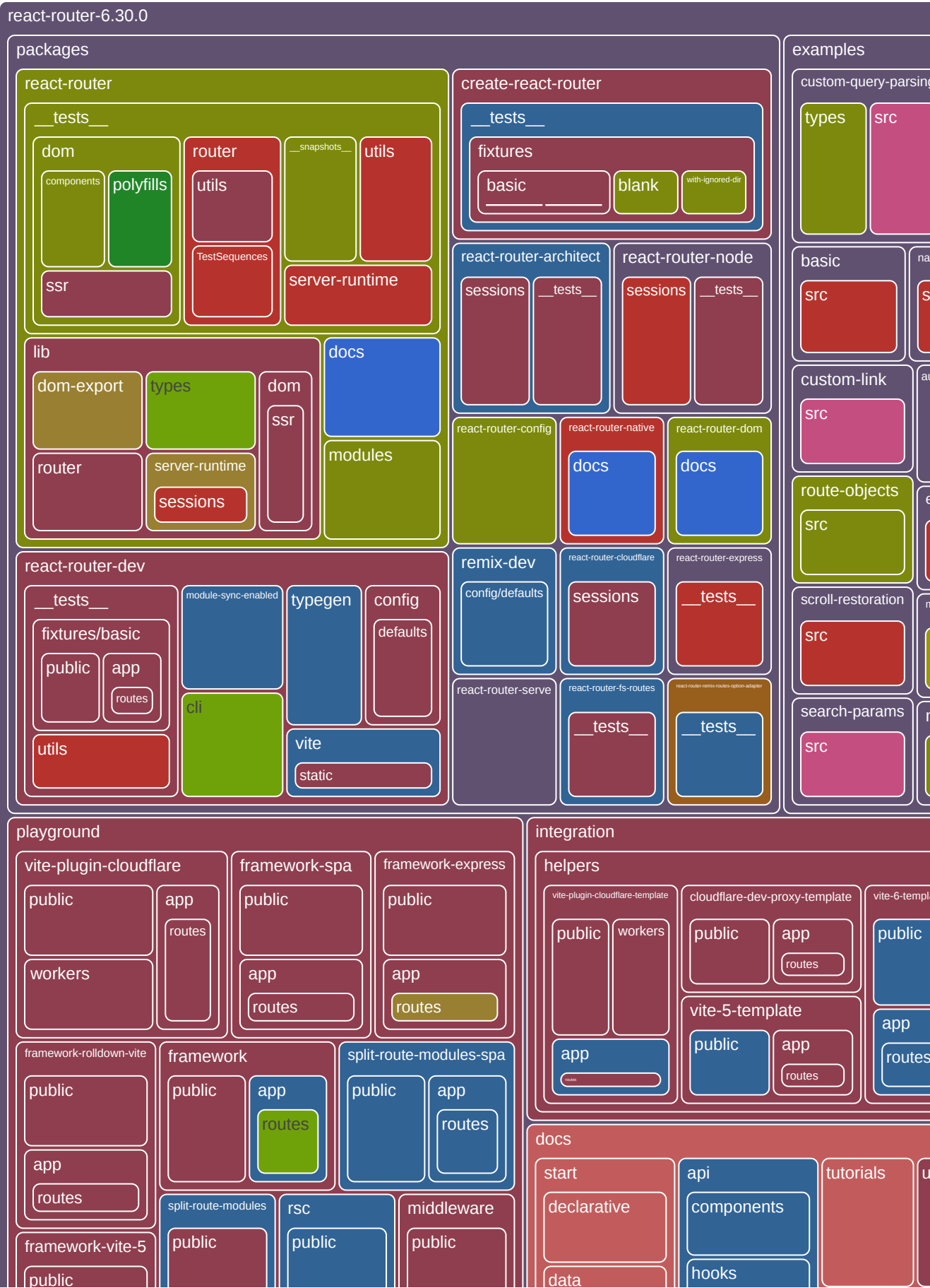
Main authors with highest number of commits

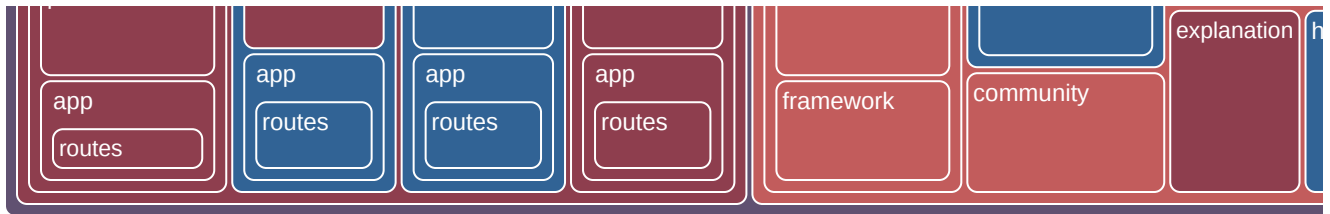




Second author per directory

Second author with the second highest number of commits





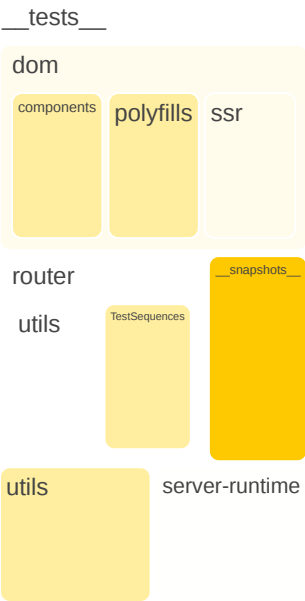
Days since last commit per directory

Days since last commit

react-router-6.30.0

packages

react-router



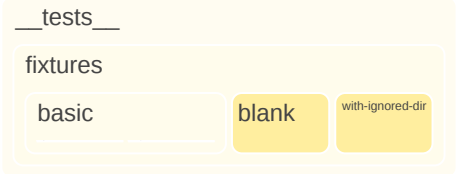
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react-router-node
sessions __tests__

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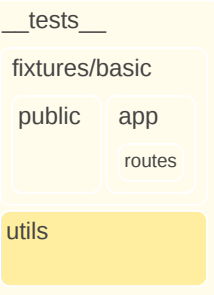
react-router-fs-routes

react-router-remix-routes-option-adaptor

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__tests__

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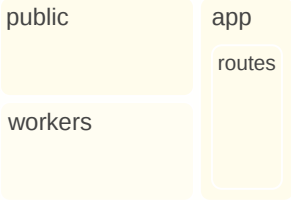
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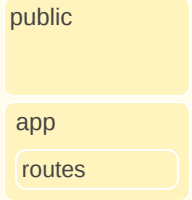
vite
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playground

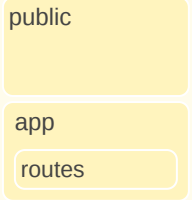
vite-plugin-cloudflare



framework-spa



framework-express



framework-rolldown-vite

public

app
routes

framework-vite-5
public

framework

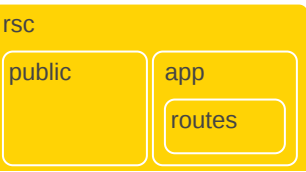
public

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split-route-modules-spa
public

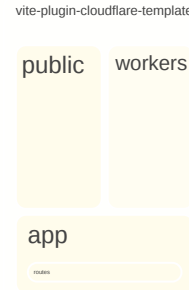
split-route-modules

public
app
routes

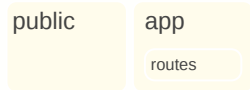


integration

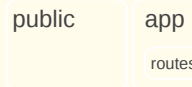
helpers



cloudflare-dev-proxy-template



vite-5-template



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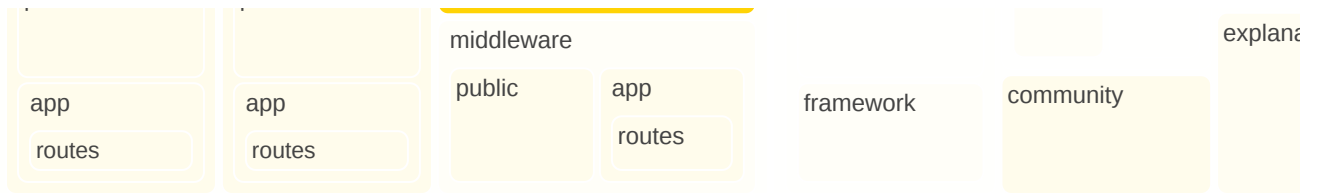
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scroll-restor

src

search-pars

src

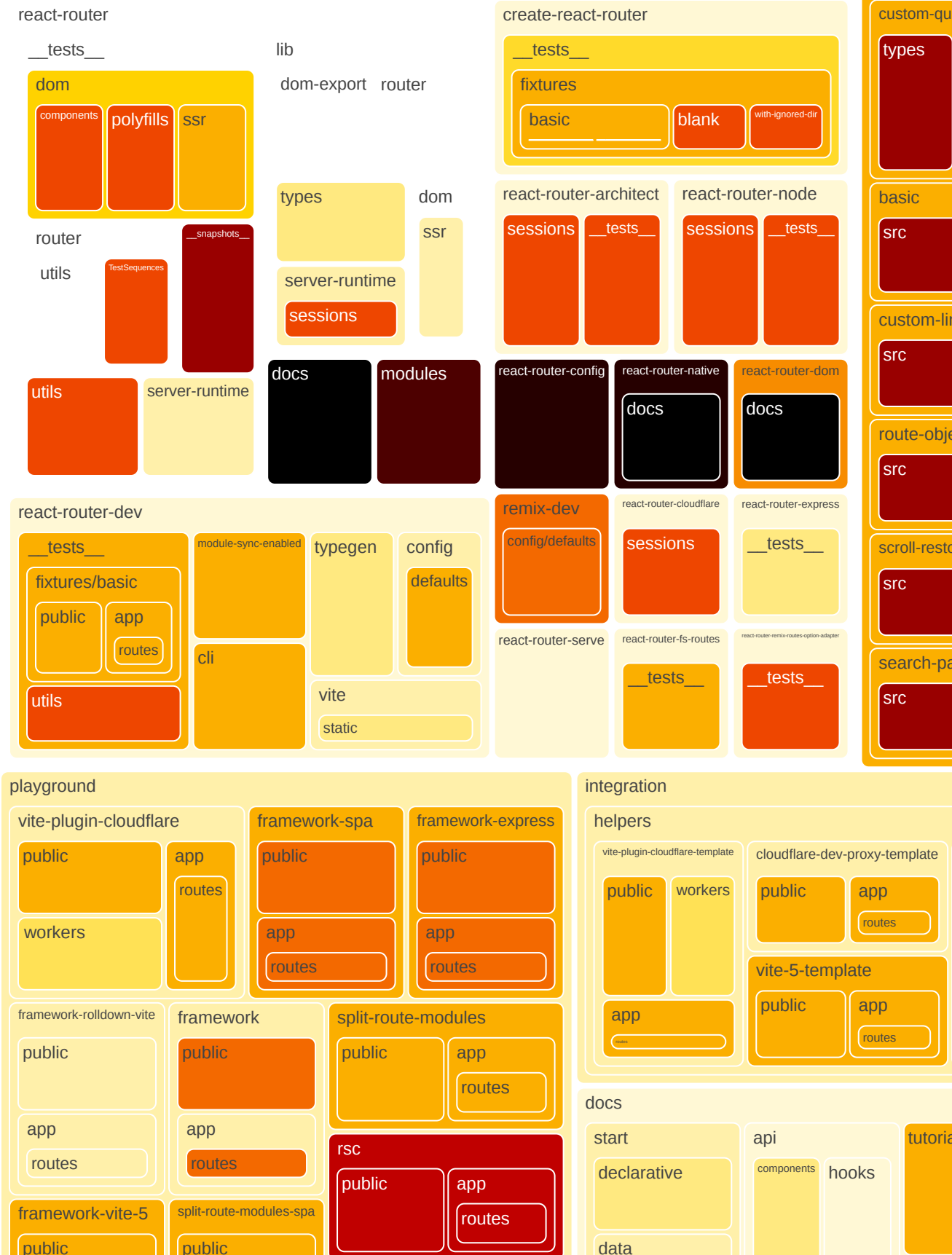


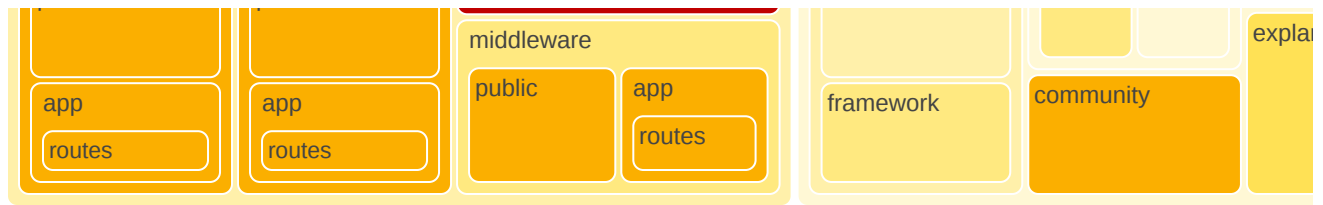
Days since last commit per directory (ranked)

Rank of days since last commit

react-router-6.30.0

packages

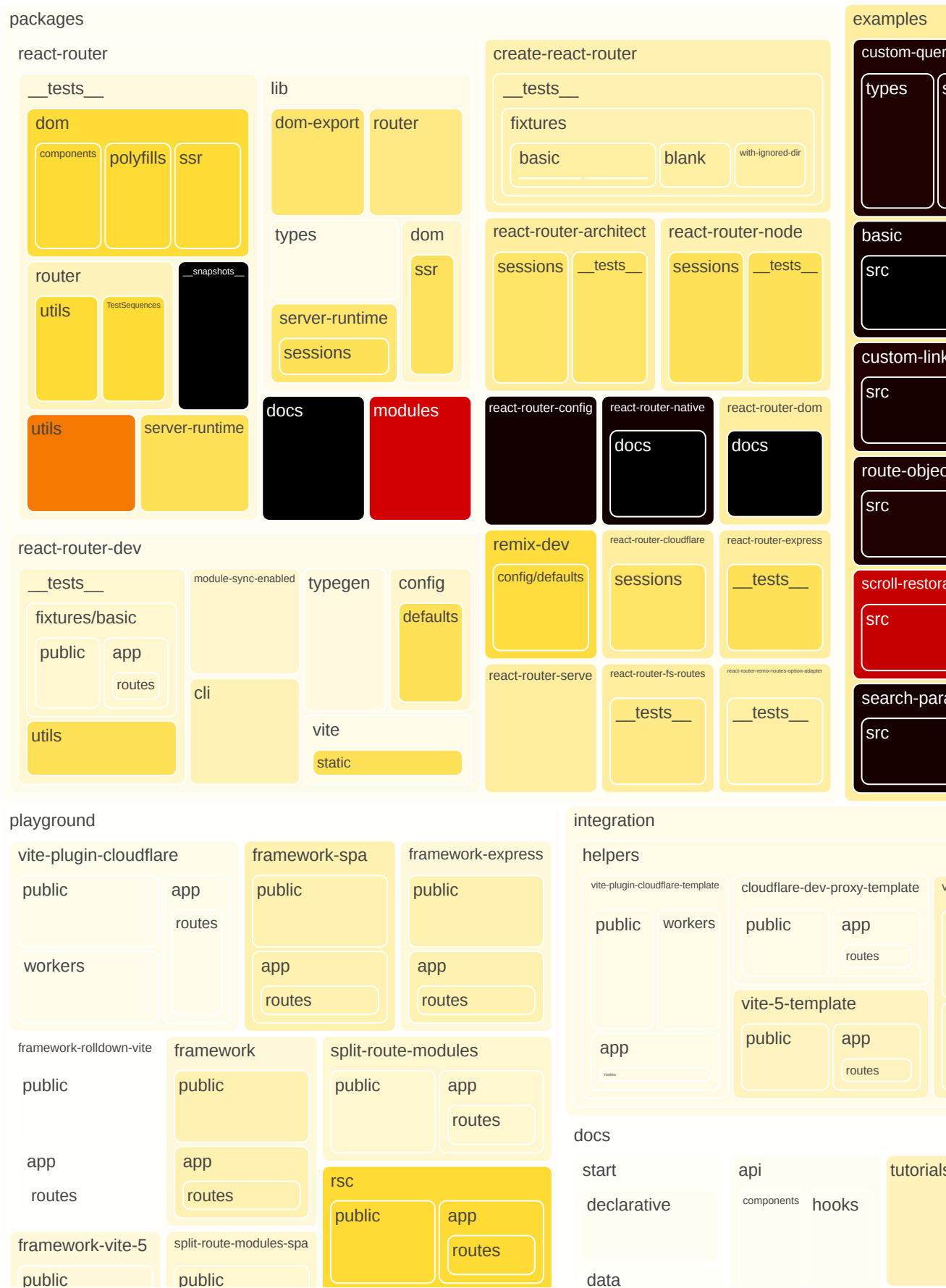


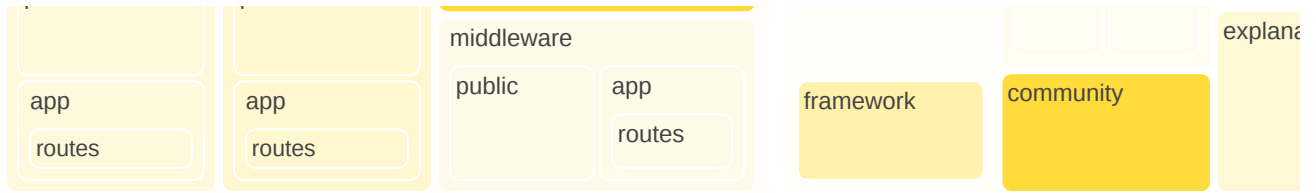


Days since last file creation per directory

Days since last file creation

react-router-6.30.0

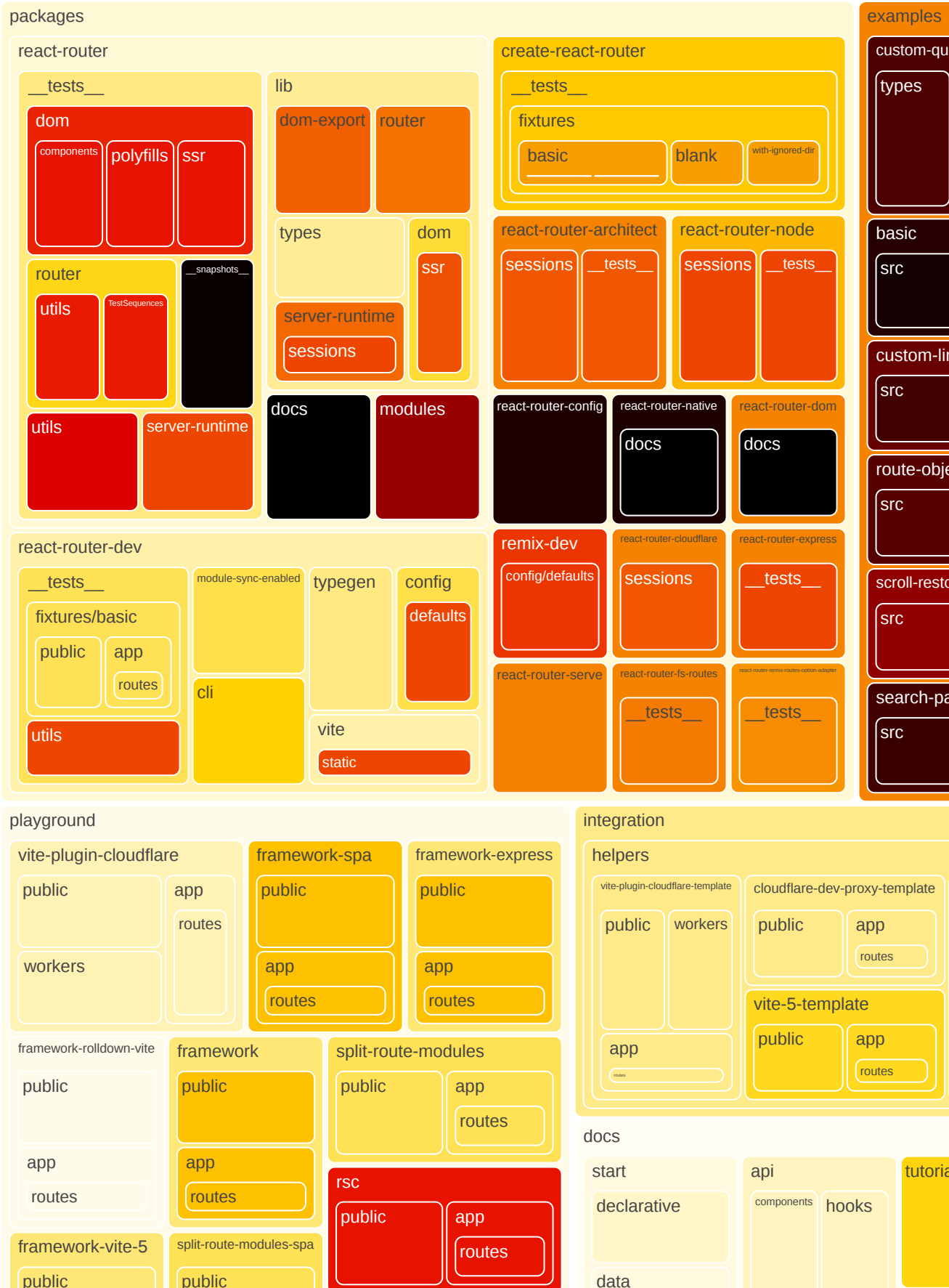


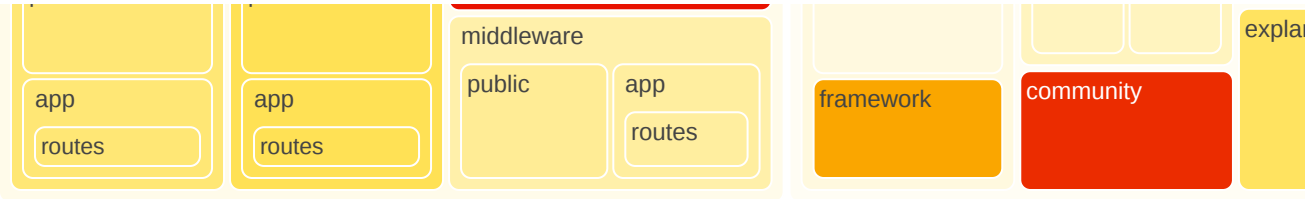


Days since last file creation per directory (ranked)

Rank of days since last file creation

react-router-6.30.0





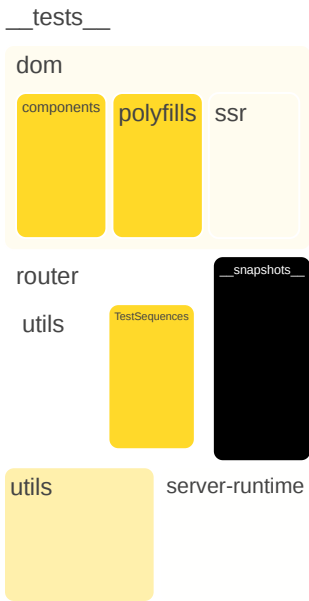
Days since last file modification per directory

Days since last file modification

react-router-6.30.0

packages

react-router



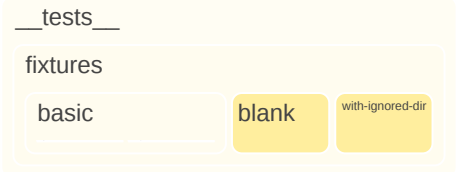
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sessions __tests__

react-router-node
sessions __tests__

react-router-config

react-router-native
docs

react-router-dom
docs

remix-dev
config/defaults

react-router-cloudflare
sessions

react-router-express
__tests__

react-router-serve

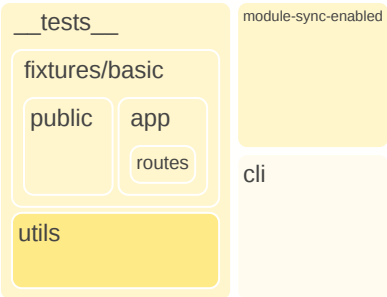
react-router-fs-routes

react-router-remix-routes-option-adapter

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react-router-dev

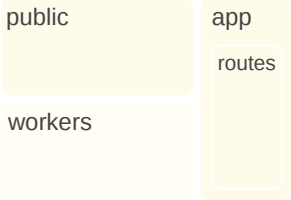


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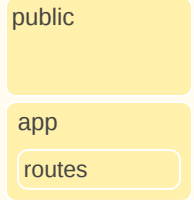
vite
static

playground

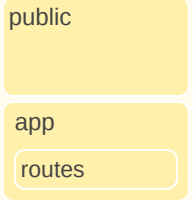
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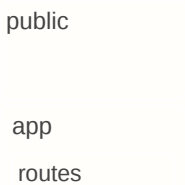
framework-spa



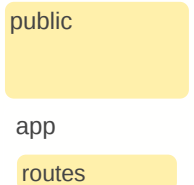
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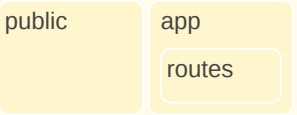
framework-rolldown-vite



framework



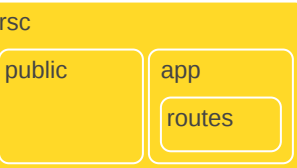
split-route-modules



framework-vite-5

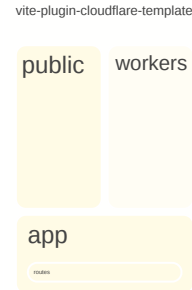


split-route-modules-spa

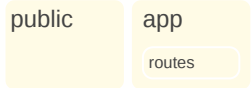


integration

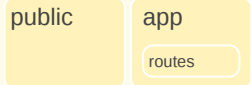
helpers



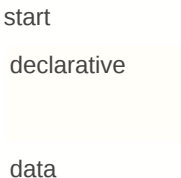
cloudflare-dev-proxy-template



vite-5-template



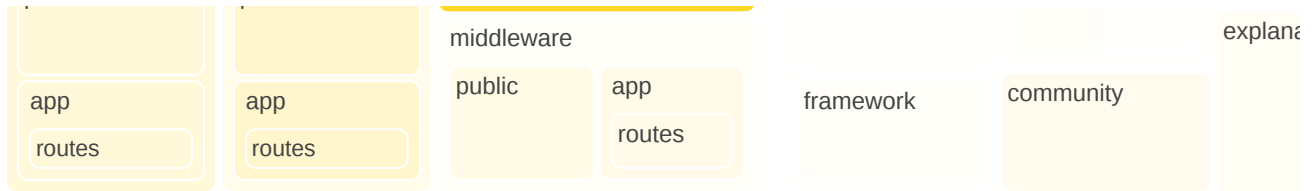
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components hooks

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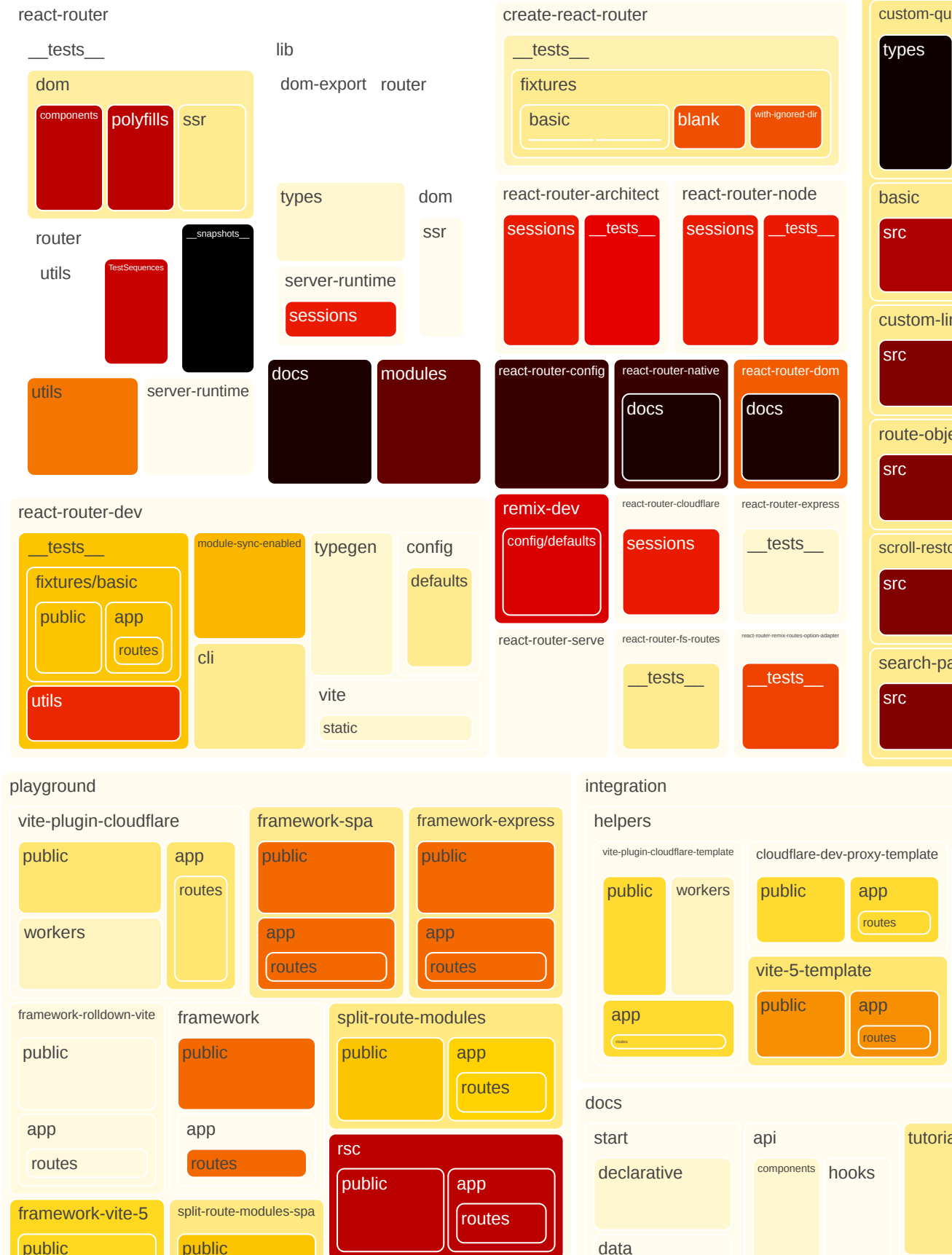


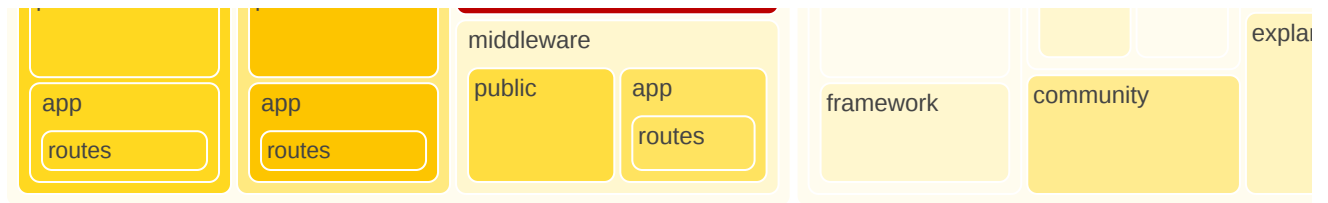
Days since last file modification per directory (ranked)

Rank of days since last file modification

react-router-6.30.0

packages

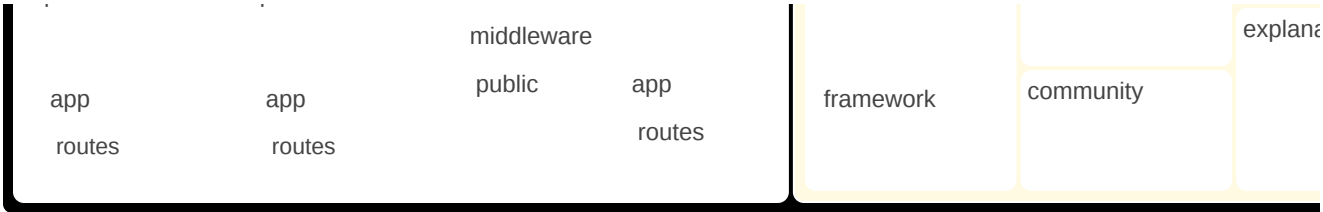




File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

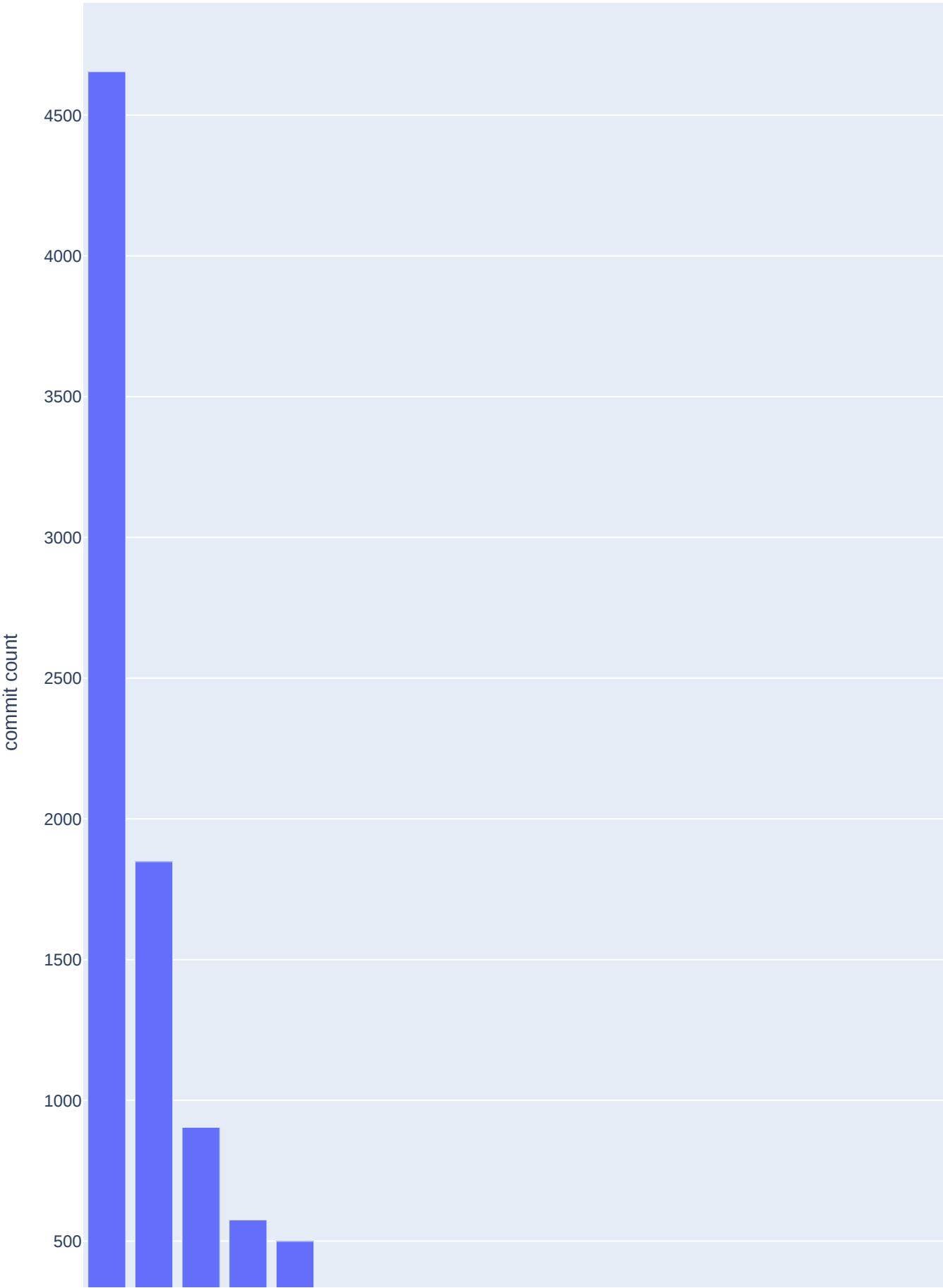
Sum of commits that changed more than 30 files (each) = 469
Max changed files with one commit = 1061

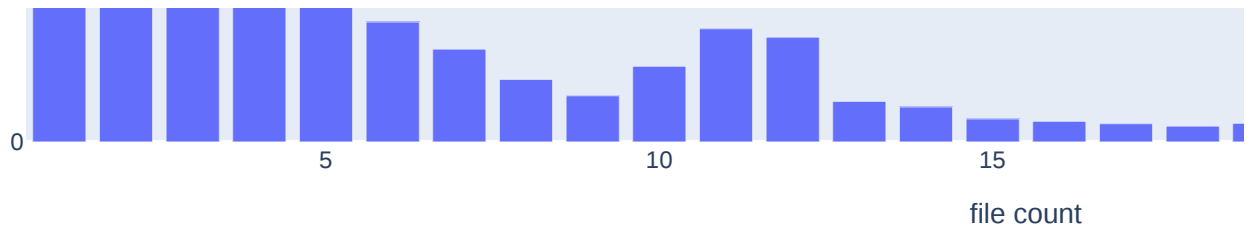
	filesPerCommit	commitCount
count	171.000000	171.000000
mean	149.959064	66.122807
std	193.597592	392.139714
min	1.000000	1.000000
25%	43.500000	1.000000
50%	87.000000	3.000000
75%	157.500000	8.000000
max	1061.000000	4654.000000

	filesPerCommit	commitCount
0	1	4654
1	2	1848
2	3	904
3	4	575
4	5	499
5	6	301
6	7	233
7	8	157
8	9	116
9	10	190
10	11	284
11	12	263
12	13	102
13	14	88
14	15	58
15	16	52
16	17	46
17	18	40
18	19	47
19	20	33
20	21	36
21	22	31
22	23	56
23	24	31
24	25	30
25	26	15
26	27	56
27	28	56
28	29	19
29	30	18

Bar chart with the number of files per commit distribution

Changed files per commit





WordCloud of git authors

	word	frequency
0	Matt Brophy	2084
1	Michael Jackson	1951
2	Ryan Florence	1307
3	Chance Strickland	511
4	Remix Run Bot	504
5	Mark Dalglish	423
6	Pedro Cattori	412
7	Tim Dorr	390
8	Jimmy Jia	389
9	Jacob Ebey	294

Wordcloud of git authors

